

RAMAIAH INSTITUTE OF TECHNOLOGY, BANGALORE – 560054

DEPARTMENT OF CHEMISTRY

SUB: ENGINEERING CHEMISTRY, CODE: CYC22, COURSE: II SEM B.E. (All branches)

CIE TEST – 2, TERM: 01-03-2024 TO 22-06-2024. MAX. MARKS: 30, TIME: 60 MIN. course outcomes: CO3, CO4 & CO5

INSTRUCTIONS: ANSWER ANY TWO FULL QUESTIONS. EACH CARRIES 15 MARKS.

1. a) What is homologous series? Describe the liquid crystalline behavior in p-azoxyanisole. (5) CO3
b) Give reasons: i) Teflon coated utensils are used for cooking or baking. (5) CO3
ii) All simple organic polymers are not monomers
c) What are nanomaterials? Mention any four applications of nanomaterials. (5) CO4
2. a) Define high polymer. Outline the synthesis and applications of Teflon (5) CO3
b) Justify: i) There is difference between a battery and fuel cell. (5) CO5
ii) A membrane is inserted close to the cathode site in methanol-oxygen fuel cell.
c) What are liquid crystals? Outline the classification of liquid crystals with example. (5) CO3
3. a) Define optical sensor. Explain the instrumentation of colorimeter. (5) CO4
b) List out any four properties and applications of organic light emitting diodes. (5) CO3
c) Describe the synthesis of nanomaterial by solution combustion method. (5) CO4